Problem 06: AI-Powered Cybersecurity Threat Detection and Attack-Chain Reconstruction
Domain: Cybersecurity | Anomaly Detection | Graph ML | Sequence Modeling
Difficulty: Hard

Problem
Attacks unfold in stages: reconnaissance, credential compromise, privilege escalation, lateral movement, exfiltration. Single alerts look harmless; threshold systems flood analysts with false positives.
Objective
Build a security analytics system that turns raw logs into prioritized, explainable incident investigations with reconstructed attack sequences.
Users
SOC analysts, incident responders.
Core Requirements

- Ingest network-flow, authentication or system logs.
- Normalize timestamps, entities, event types.
- Behavioral baselines per user/device/endpoint.
- Detect suspicious events (supervised or unsupervised ML).
- Correlate events via time windows and entity relations.
- Severity scoring of incidents.
- Reconstruct probable attack sequences.
- Explain contributing events.
- Investigation dashboard with filters and timelines.
- Evaluate detection quality, false positives, latency.

Advanced Challenges

- Anomaly detection + sequence classification together.
- Graph representation of events and entities.
- Unseen attack patterns.
- Severe class imbalance.
- Alert-fatigue reduction by correlation.
- Time-based splits to avoid leakage.
- Measuring attack-chain reconstruction quality.

Suggested Datasets
CICIDS2017, UNSW-NB15, LANL authentication dataset, CERT insider threat, DARPA OpTC; map findings to MITRE ATT&CK.
Suggested Tech
Python, scikit-learn, XGBoost, Isolation Forest, PyTorch (LSTM/Transformer, GNN), NetworkX, OpenSearch, FastAPI.
Reference Architecture
Log ingest → normalizer → feature store (windowed aggregates) → detectors (IF + supervised) → entity graph → correlator → chain builder (ATT&CK stage ordering) → scorer → dashboard.
Evaluation
Precision, recall, PR-AUC, false positives per day, detection latency, chain-reconstruction F1 vs labelled campaigns, alert reduction ratio from correlation.
Expected Deliverable
SOC dashboard: anomalies, prioritized incidents, related entities, timelines, evidence, performance stats.
Scope Note

Use authorized, synthetic or public datasets. Defensive monitoring only.

Pitfalls

Random split leakage in time-ordered logs; reporting accuracy on imbalanced data.

Suggested Timeline (48h)

0-8h ingest + features; 8-20h detectors; 20-32h correlation + graph; 32-40h chain reconstruction; 40-48h dashboard + eval.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work